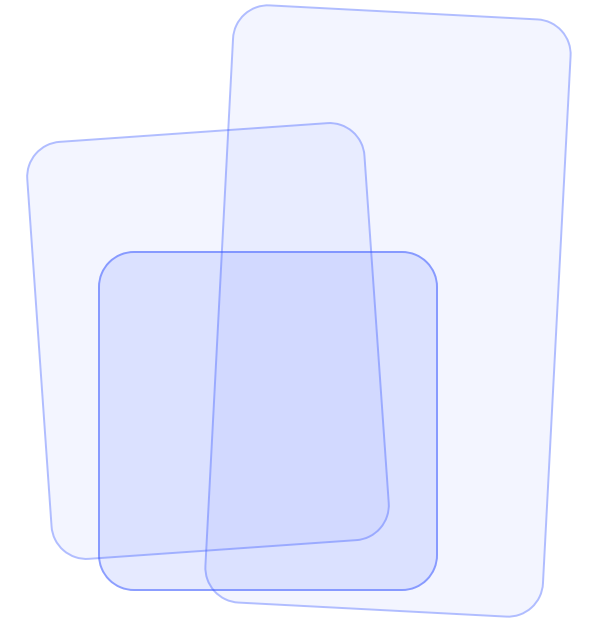


Clinical Trial Risk Monitor & Protocol Deviation Detector

•

Real-time compliance monitoring and explainable CAPA generation copilot



TEAM

DOMinators

PRODUCT TAG

**Continuous Audit
Readiness via AI**

Clinical trial risk compounds while audits wait

QUARTERLY AUDIT LAG

Weeks

to detect deviations through formal audit cycles.
Manual monitoring cannot keep pace with trial activity.

ESCALATING FINANCIAL RISK

\$50M–\$100M

A single trial rejection can delay market approval by 6–12 months.
Cost of failure compounds while exposure stays hidden.

EXPOSURE	WHAT STAYS UNSEEN
Scale dilemma	5,000+ patient visits across 200+ sites make manual monitoring impossible.
Hidden exposure	Missed visits, dosing errors, and banned co-medications can remain unseen until formal FDA audits.

Continuous monitoring turns hidden deviations into ranked action

1

01 · INPUT

Protocol + Visits

Patient visit records evaluated continuously against protocol specifications.

2

02 · DETECT

Deviation Engine

Classify Major, Minor, or Administrative under ICH E6(R2) GCP.

3

03 · RANK

Site Risk Ranking

Dynamic 0–100 site risk index from leading indicators.

4

04 · ACT

Automated CAPA

Evidence flow and explainability stay attached to the action.

Instant detection · Regulatory classification · Transparent risk scoring

The architecture separates deterministic checks from ambiguous clinical logic

INGESTION

Protocol rules

Parsed into structured specifications

Visit logs

PostgreSQL + FAISS/Chroma vector store

DETECTION & SCORING

Deterministic rule layer

Checks explicit protocol specifications

RAG

Resolves ambiguous clinical logic with context

ORCHESTRATION

FastAPI orchestrator

React drill-down dashboard: Trial → Site → Patient → Visit

OUTPUTS

Risk outputs

Deviation classification + site risk index

CAPA generation

Evidence-backed corrective and preventive action

One investigation path ends in an audit-ready CAPA

01 RANK

Trial Overview

Surface the highest-risk investigation centers from the trial overview.

Site ranking

02 EXPLAIN

Site Drill-down

Drill into recency, repeat offenses, severity mix, and clause-level protocol evidence.

Risk drivers + violated section

03 ACT

Audit-Ready CAPA

Generate Root Cause, Corrective Action, and Preventive Action in one click.

Evidence attached

LINKED REPORT OUTPUT

Violated protocol section called out beside the generated CAPA record.

IBM technologies anchor inference and codebase integrity

INFERENCE

watsonx.ai

Powers RAG-grounded severity rationale and context-aware CAPA drafting.

Clinical logic · explanation · drafting

CODEBASE INTEGRITY

IBM Bob

Pipeline assembly. Assembles the multi-module orchestration pipeline in [orchestrator.py](#).

Contract consistency. Scans the repository to reconcile FastAPI endpoint schemas with API contracts.

The result is faster detection with evidence attached

DETECTION RECALL

≥95%

Validated on synthetic test cases with injected protocol violations.

Automated pipeline

CAPA VELOCITY

<30 sec

CAPA generation drops from weeks of manual investigation to under 30 seconds.

Audit-ready output

CITATION COVERAGE

100%

Generated CAPAs and severities carry direct citations to protocol clauses.

Zero-hallucination tolerance

Measurement context: synthetic clinical test cases · injected protocol violations · direct protocol-clause citations

DOMinators combine compliance logic, analytics, and product delivery

DATA/AI LEAD 1

Deviation detection

Deviation detection engine and ICH E6(R2) severity classification rules.

Priyansh Balwani

DATA/AI LEAD 2

Risk scoring

Site-level risk scoring model and trend projection analytics.

Arya Kayastha

DATA/AI LEAD 3

CAPA pipeline

Synthetic clinical data generation and RAG-driven CAPA pipeline.

Jay Changanani

FULL-STACK DEVELOPER

Product delivery

FastAPI backend gateway, React dashboard, and IBM Bob agent integration.

Nancy Vaghela